

Anthony Schurle

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Research Interests

Logic in Computer Science; Descriptive Complexity; Finite Model Theory

Education

Rice University	Houston, Texas
<i>B.S. Mathematics, B.A. Computer Science</i>	Aug. 2024 – May 2028
GPA: 3.8/4.00	
Northland Christian School, Texas	Aug. 2020 – May 2024
<i>Valedictorian</i>	

Honors & Awards

Trustee Distinguished Academic Scholarship (\$80,000), Rice University	Apr. 2024
President's Honor Roll , Rice University	Fall 2025

Relevant Coursework

Graduate & Cross-Listed: General Topology, Automata/Formal Languages/Computing
Reading Course: Tangle-tree duality in abstract separation systems
Undergraduate: Abstract Algebra I, Honors Analysis, Reasoning About Algorithms, Number Theory, Graph Theory, Honors Linear Algebra, Honors ODEs, Honors Statistics
Planned: Logic In Computer Science, Abstract Algebra II, Complex Analysis, Probability Theory
Independent Study: *An Introduction to Mathematical Logic* (Enderton), *How to Prove It* (Velleman)

Professional Experience

Teaching Assistant <i>Computational Thinking (COMP140)</i>	Aug. 2026 – Dec. 2026
George R. Brown School of Engineering and Computing, Rice University	
Teaching Assistant <i>Algorithmic Thinking (COMP182)</i>	Jan. 2026 – May 2026
George R. Brown School of Engineering and Computing, Rice University	
Teaching Assistant <i>Computational Thinking (COMP140)</i>	Aug. 2025 – Dec. 2025
George R. Brown School of Engineering and Computing, Rice University	

Projects

Graft (Interactive Graph Theory Sandbox) | *JavaScript, SVG*

- Developed a Desmos-style web application for building and exploring graphs through an intuitive visual interface
- Implemented real-time computation of graph properties (degree sequences, connectivity, bipartiteness, chromatic number)
- Created export functionality for set notation, adjacency matrices, and LaTeX TikZ code

r2a (Disassembler) | *Python, RISC-V*

- Developed a formal specification for RISC-V instruction decoding using decision-tree optimization
- Implemented a testing pipeline to verify binary equivalence

Epidemiological Transmission Mapping | *Python*

- Modeled bacterial infection spread as a complete weighted directed graph using genetic (Hamming distance) and epidemiological data
- Implemented greedy algorithms to infer a rooted directed minimum spanning tree (RDMST) representing likely transmission pathways

Technical Skills

Languages: Python, C, Java, JavaScript, RISC-V Assembly, x86-64 Assembly

Tools: LaTeX, Git & GitHub, Docker, Linux